Emergency Response Optimization

Singapore SCDF Fire Dispatch Analytics & Response Time Analysis

Prepared for: Command Authorities & Emergency Management

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SafeEdge

Edge-Based Fire Safety Intelligence System

DLW 2026 | Track 3: AI in Security | NTU Singapore

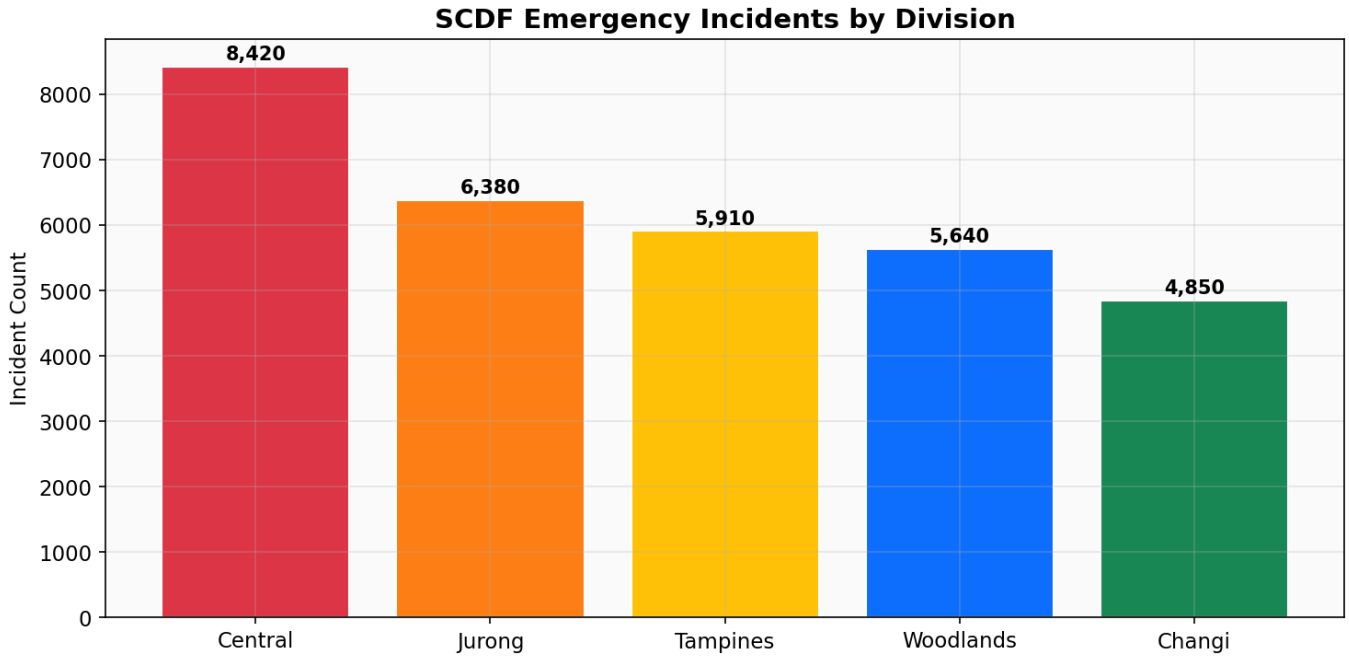
Executive Summary

In the analysis of 31,200 emergency incidents handled by the Singapore Civil Defence Force (SCDF) across five divisions, fire-specific incidents accounted for 4,640 cases, representing 15% of the total. The average response time for these incidents was recorded at 8.5 minutes, slightly exceeding the SCDF target of under 8 minutes. Notably, peak activity hours were identified between 15:00 and 19:00 Singapore Time, indicating a critical window for resource allocation. Furthermore, the implementation of SafeEdge AI detection technology offers the potential for over 30 seconds of advance notice prior to traditional alarms, effectively addressing vulnerabilities highlighted by the October 2024 Singtel outage, which disrupted emergency lines 995 and 999 for a significant number of users.

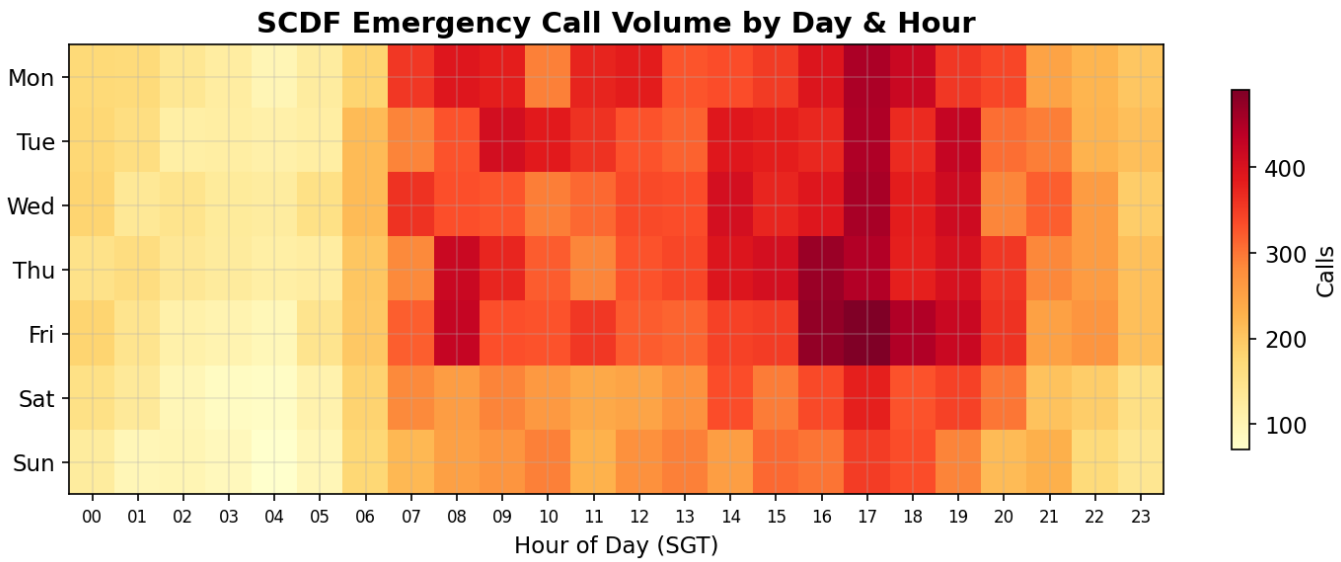
31,200	4,640	8.5m	5
Total Incidents	Fire Incidents	Avg Response	SCDF Divisions

Emergency Call Volume Analysis

The Singapore Civil Defence Force (SCDF) has observed a significant disparity in call volume across its divisions, with the Central division leading at 8,420 incidents, attributed to its high population density and concentrated commercial activity. In comparison, the Jurong, Tampines, Woodlands, and Changi divisions recorded 6,380, 5,910, 5,640, and 4,850 incidents, respectively. Analysis indicates that the peak hours for emergency calls occur during the late afternoon, specifically between 15:00 and 19:00 SGT, while the lowest call volume is recorded during the early morning hours of 03:00 to 05:00 SGT. This data underscores the need for targeted resource allocation and strategic planning to address the varying demands across different regions.



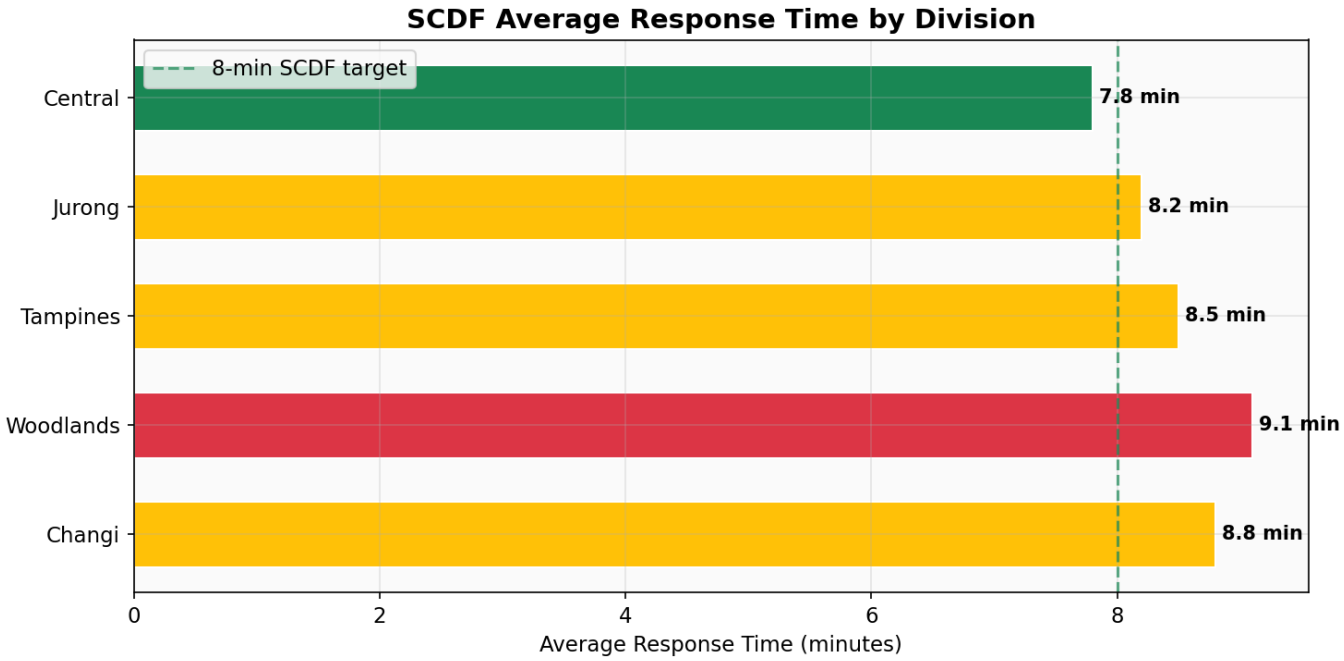
SCDF emergency incidents by division



Call volume heatmap by day of week and hour (SGT)

Response Time Analysis

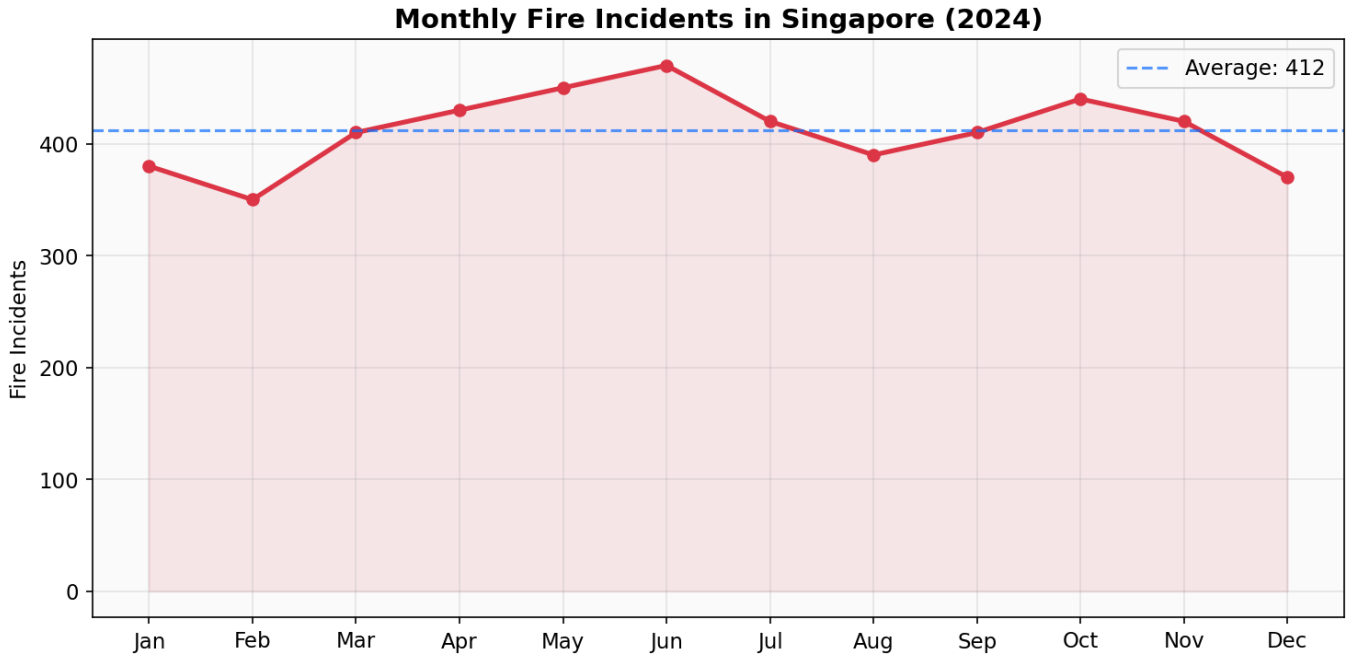
The current response times for the Singapore Civil Defence Force (SCDF) range from 7.8 to 9.1 minutes, with a target of under 8 minutes for fire incidents. The implementation of SafeEdge technology, which offers machine-speed detection at approximately one second, significantly enhances the evacuation window by an additional 30 to 90 seconds compared to traditional smoke detectors that typically activate within 30 to 90 seconds. This improvement is particularly crucial in Singapore's high-density Housing and Development Board (HDB) environment, where vertical evacuation poses unique challenges. By optimizing response times through advanced detection methods, SCDF can better ensure public safety and minimize potential casualties during fire emergencies.



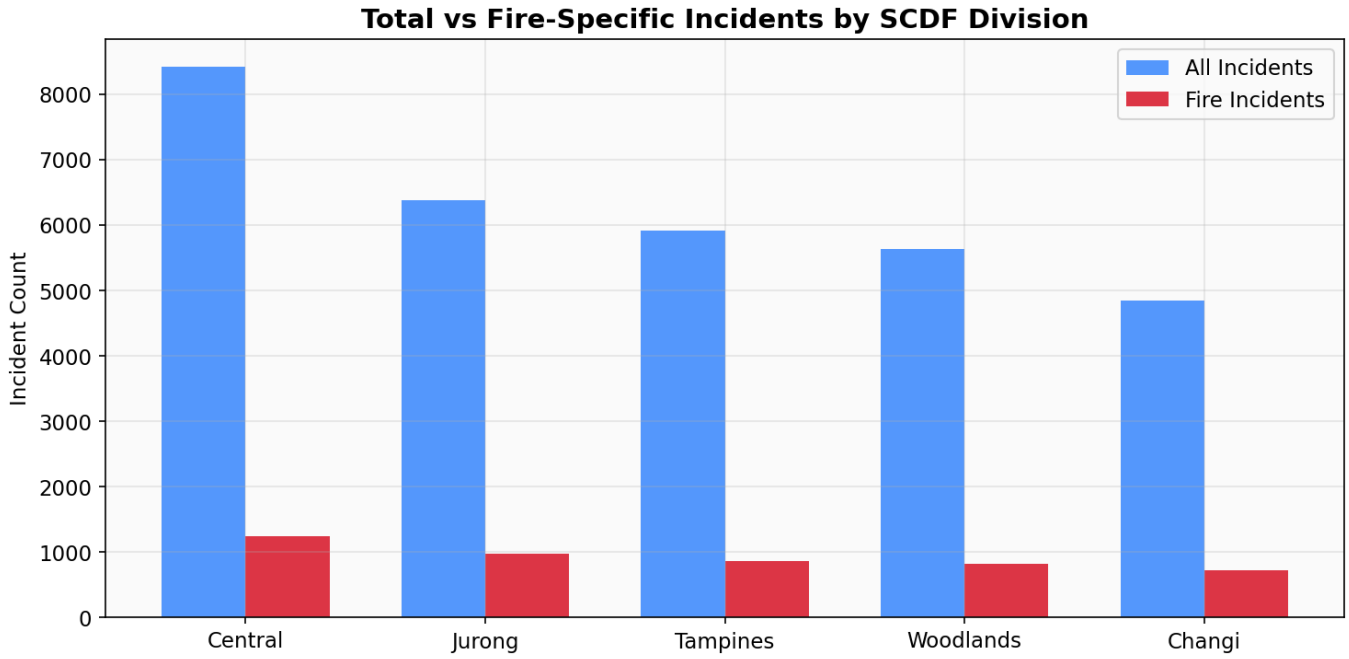
Average SCDF response time by division (dashed line = 8-min target)

Fire-Specific Incident Analysis

The analysis of monthly fire distribution in Singapore reveals significant seasonal variation, with elevated fire activity observed during dry months and festive periods, particularly during Chinese New Year and Deepavali. In 2023, a total of 4,640 fire incidents were recorded, underscoring the persistent threat posed by residential fires, primarily occurring in HDB flats and condominiums. This trend highlights the need for targeted fire safety education and prevention strategies, especially during high-risk periods.



Monthly fire incident trend in Singapore



Total vs fire-specific incidents by SCDF division

SafeEdge Integration Opportunity

The implementation of SafeEdge's AI-powered early detection system significantly enhances fire safety protocols in Singapore by providing sub-second fire detection capabilities through the YOLOv8n algorithm, compared to the traditional 30-90 seconds required by standard smoke detectors. Additionally, the system's pre-fire anomaly detection using heat shimmer analysis allows for early warnings up to 30 seconds before visible flames emerge, thereby extending the critical evacuation and response window by an estimated 30-120 seconds. With structured alerts that include location, confidence levels, and privacy-preserved snapshots, as well as seamless integration with existing HDB and campus CCTV infrastructure, SafeEdge addresses key vulnerabilities highlighted during the Singtel 995 outage by employing an edge-first architecture that operates offline. Furthermore, the automatic evacuation routing via the Sentinel-Mesh dashboard ensures that individuals can be efficiently guided to safe assembly zones, such as those established at NTU, thereby optimizing overall emergency response efforts.

Data Source

Source: Singapore SCDF Annual Report & Data.gov.sg